

VIETNAM NATIONAL UNIVERSITY OF HOCHIMINH CITY
THE INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



An LLM-Augmented Approach for Topic Modeling and Emotion Recognition in Vietnamese Online Education Platforms

By
Nguyễn Gia Huy

A thesis submitted to the School of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
Bachelor of Information Technology/Computer Science/Computer Engineering

Ho Chi Minh City, Vietnam
2025

An LLM-Augmented Approach for Topic Modeling and Emotion Recognition in Vietnamese Online Education Platforms

APPROVED BY:

_____,
Nguyen Van A, Ph.D, Chair (*Example*)
*(Type Committee names beneath lines)

(Typed Committee name here)

(Typed Committee name here)

(Typed Committee name here)

(Typed Committee name here)

THESIS COMMITTEE
(Whichever applies)

ACKNOWLEDGMENTS

It is with deep gratitude and appreciation that I acknowledge the professional guidance of Dr. Nguyễn Thị Thanh Sang. Her expertise and insights have been instrumental in guiding my work, and I am particularly grateful for all the constructive feedback and invaluable knowledge she has provided.

I extend my gratitude to my friends for their unconditional assistance throughout this research. Furthermore, I wish to express my deep appreciation to my family for their unwavering motivation, belief, and support on this journey. Their constant encouragement has provided the strength necessary for me to achieve this goal.

Note: Paper A4, Top: 2.5cm; Bottom: 2cm; Left: 3cm; Right: 2cm; double spaced, and 12 point font

TABLE OF CONTENTS

ACKNOWLEDGMENTS	ii
TABLE OF CONTENTS	iii
LIST OF FIGURES	v
LIST OF TABLES	vi
ABBREVIATIONS	vii
ABSTRACT	viii
CHAPTER 1	1
INTRODUCTION	1
1.1. Background	1
1.2. Problem Statement	1
1.3. Scope and Objectives	1
1.4. Assumption and Solution	1
1.5. Structure of thesis	1
CHAPTER 2	2
LITERATURE REVIEW/RELATED WORK	2
2.1. Review 1	2
2.1.1. Sub review 1	2
2.2. Review 2	2
2.2.1. Sub review 2	2
CHAPTER 3	3
IMPLEMENTATION	3
3.1. User requirement analysis	3
3.1.1. Sub 1	3
3.2. System Design	3
3.2.1. Database design	3
3.2.2. User Interface design	3
3.3. Abc	3

CHAPTER 4	4
EXPERIMENTS AND RESULTS	4
4.1. Experiments	4
4.1.1. Abc	4
4.2. Results	4
4.2.1. Result 1	4
4.2.2. Result 2	4
CHAPTER 5	5
DISCUSSION AND EVALUATION	5
5.1. Discussion	5
5.2. Comparison	5
5.3. Evaluation	5
CHAPTER 6	6
CONCLUSION AND FUTURE WORK	6
6.1. Conclusion	6
6.2. Future work	6
REFERENCES	7
APPENDICES	8

LIST OF FIGURES

Figure 1.1: Abc

LIST OF TABLES

Table 3.1: Cde

ABBREVIATIONS

IT: Information Technology

...

// Make sure your abbreviation list is in alphabetical order.

ABSTRACT

Abc...

// This page should be double spaced with a similar format to the rest of the document.

CHAPTER 1

INTRODUCTION

1.1. Background

The past few years have witnessed a drastic advancement in technology, enhancing nearly every aspect of life. Education, in particular, is a field that has seen a significant shift toward online learning. For this reason, the need for a system that can understand student feedback and concerns has become increasingly important for the education system. Monitoring discussion forums and analyzing student feedback and comments is needed for gaining valuable insights, which can be used to tailor the teaching and management of the education system.

1. Challenges in Vietnamese Natural Language Processing (NLP)

Vietnamese is considered a low-resource language in natural language processing (NLP) research[1], making the implementation of language analysis systems such as sentiment analysis and topic modeling quite a difficult task. Historically, research on Vietnamese sentiment analysis (SA) has been hampered by the limited availability of suitable datasets[2]. Existing educational datasets for Vietnamese NLP often lack sufficient domain relevance and student slang, which presents a significant challenge for conducting research and improving the educational experience.

2. The demand for Advanced Text Analysis in Education

The two core methods of Text Analysis are Topic Modeling and Sentiment Analysis. These are essential for extracting meaning from the enormous volume of student-generated content, such as feedback, posts, and comments on online platforms.

Emotion Recognition (ER): is a higher-level approach than sentiment analysis[3]. In ER, text is not classified into general sentiments (such as positive, negative, or neutral) but into fine-grained emotions like joy, sadness, etc[3]. This detailed perspective will help gain a deeper understanding of student discussions. The foundational corpus for this work is the UIT-VSMEC (Vietnamese Social Media Emotion Corpus), which contains 6,927 human-annotated sentences across seven emotion labels[3].

Topic Modeling (TM) for context extraction: Topic modeling is a method for extracting “topics” from documents. However, in this research, this technique will serve a supporting role. The purpose is to extract and organize underlying themes from the student-generated content, which is necessary for interpreting and validating the detected emotions. By analyzing both the topics and the emotions, complex student feelings can be contextualized for a better understanding.

LLM Data Augmented Approach: The advancement of LLMs has brought up a new way to generate and labeling data. Many recent researches show that LLM has the capability to perform data diversity and generating by paraphrasing, response synthesis, counterfactuals, and many more[4]. This can be done by instructing LLM to generate synthetic examples exhibiting a given label by providing a labeled example and prompting the LLM to generate similar content[5].

1.2. Problem Statement

Implementing a reliable Fine-Grained Emotion Recognition for Vietnamese education context encounters several difficulties that can be solved by the LLM-Augmented approach:

1. **Poor performance and robustness in Fine-grained emotion**

recognition task: BERT models achieve very high accuracy for basic sentiment polarity (e.g., positive, negative, neutral)[]. However, their performance drops noticeably when trying to classify documents into more detailed human emotions. For instance, the best classification model using CNN+word2vec on the UIT-VSMEC corpus only managed to achieve a weighted-F1 score of 59.74%[].

2. **Data scarcity and imbalanced dataset:**

As mentioned before, Vietnamese is a low-resource language, hence domain-specific datasets are scarce[]. The dataset used in this research, UIT-VSMEC, suffers from a heavy imbalance in data distribution, as the "other" label occurs most frequently, and other emotions are lacking[]. This limits the model's robustness.

1.3. Scope of research and Objectives

Scope of research:

This study focuses on analyzing Vietnamese student-generated text from online platforms. The research specifically addresses two core NLP tasks:

1. **Fine-Grained Emotion Recognition:**

The primary classification task involves identifying detailed emotions (e.g., enjoyment, sadness, fear, anger, disgust, surprise) within documents or text data.

2. **Topic Modeling:**

The supporting task involves identifying and extracting a coherent thematic structure (topic) from the text to provide contextual

support for the emotion analysis.

Objectives:

The objectives of this research are:

1. To leverage LLMs for controlled data augmentation to enrich and balance low-resource emotions.
2. To construct an LLM-BERT ensemble framework where the LLM acts as the selector to improve reliability and reduce misclassification. And as a result improve the accuracy of predictions.
3. To extract topics from text using LLMs to provide supporting context for the emotion extracted from text.
4. To evaluate the combined impact of augmentation and LLM-BERT ensemble on model performance across the dataset.

1.4. Assumption and Solution

Abc...

1.5. Structure of thesis

Abc...

CHAPTER 2

LITERATURE REVIEW/RELATED WORK

2.1. Review 1

The challenges encountered by Vietnamese NLP

2.1.1. Sub review 1

Data Scarcity and Domain relevance

As a low-resource language, Vietnamese suffers from a scarcity of reliable NLP datasets. Existing datasets, such as UIT-VSFC, VLSP-SA, NEU-ESC, and ViHSD, offer only basic sentiment labels and do not cover fine-grained emotion. While the UIT-VSMEC dataset does include fine-grained emotion, it is restricted to social media data and is not specialized for the education domain. This limitation significantly hinders research in Vietnamese NLP, particularly in the educational sector.

2.2. Review 2

Sentiment Analysis (SA) and Emotion Recognition (ER) for Vietnamese Social Media Text.

2.2.1. Sub review 1

Vietnamese NLP witnessed a shift to transformer-based models, particularly BERT. Initial research with multi-lingual BERT models outperformed traditional proposed methods such as SVM and CNN, providing a stronger baseline. However, monolingual BERT models like PhoBERT, VisoBERT, and CafeBERT consistently

produce better results than multi-lingual BERT due to being trained on a large Vietnamese corpus and using optimized tokenizers.

Despite possessing relatively strong results, the current Vietnamese monolingual BERT models present limitations:

- Lack of quality emotion-focused datasets; the available data has imbalance issues.
- Perform poorly on fine-grained emotion classifying task.

2.3. Review 3

Data Augmentation by LLMs

2.3.1. Sub review 1

The work by Solomon Ubani et al. (2023) directly supports the strategy of leveraging LLMs for augmenting low-resource datasets. This method performs significantly better than traditional data augmentation techniques such as Easy Data Augmentation (EDA), Back Translation, CBERT, and BART-based methods, particularly in low-resource settings.

CHAPTER 3

IMPLEMENTATION

3.1. User requirement analysis

Abc...

3.1.1. Sub 1

Abc...

3.2. System Design

Abc....

3.2.1. Database design

3.2.2. User Interface design

3.3. Abc

CHAPTER 4

EXPERIMENTS AND RESULTS

4.1. Experiments

The initial step of experiment is to train and test models with unaugmented data for a baseline model.

4.1.1. PhoBERT-Large with unaugmented data

PhoBERT-Large with 370 millions parameters was used to train with the initial data. This generates the initial result, which is used as the baseline.

4.1.2. VisoBERT with unaugmented data

VisoBERT was used to train with the initial data. This generates the initial result, which is used as the baseline.

4.1.3. CafeBert with unaugmented data

CafeBERT was used to train with the initial data. This generates the initial result, which is used as the baseline.

4.1.4. XML-Roberta with augmented data

XML-Roberta was used to train with the initial data. This generates the initial result, which is used as the baseline

4.2. Results

The baseline results are BERT models train and test with unaugmented data

4.2.1. Result 1

Model	Accuracy	MacroF1
PhoBERT-Large	0.591837	0.562221
VisoBERT	0.634111	0.600481
CafeBERT	0.626822	0.598685
XML-Roberta	0.501458	0.506959

4.2.2. Result 2

Abc...

CHAPTER 5

DISCUSSION AND EVALUATION

5.1. Discussion

Abc...

5.2. Comparison

Abc...

5.3. Evaluation

Abc...

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1. Conclusion

Abc...

6.2. Future work

Abc...

REFERENCES

1. MARTY HALL, LARRY BROWN “Core Servlets and JavaServer Pages™ - Volume 1 Core Technologies”, Second Edition, 2003.
2. MARTY HALL, LARRY BROWN “Core Web Programming”, Second Edition, 2001.
3. DAVE CRANE, ERIC PASCARILLO, DARREN JAMES, “Ajax in Action”, 2006.
4. JAMES L. WEAVER, KEVIN MUKHAR, AND JIM CRUME, “Beginning J2EE 1.4: From Novice to Professional”, 2004.
5. ?

APPENDICES

